
1 VERTEX IDENTITIES

1.1 Massless boson polarization

In Quantum Electrodynamics (QED), we have the interaction Lagrangian.

$$H_{\text{int}} = e \int d^4z \bar{\psi} \gamma^\mu \psi A_\mu$$

where $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$ is left invariant if J^μ is conserved. We perform a Lorenz gauge fixing on the coupling of the current J^μ to the field A^μ

$$\partial^\mu A_\mu = 0. \tag{1}$$

Let us consider the plane wave field ansatz

$$A_\mu = \epsilon_\mu e^{-ip \cdot x}$$

where ϵ_μ is the polarization vector and p^μ is the four-momentum. Then, we can take the derivative

$$\partial_\mu A^\mu = -i \epsilon_\mu p^\mu e^{-ip \cdot x}$$

which satisfies the gauge fixing condition (1) when

$$\epsilon_\mu p^\mu = \epsilon \cdot p = 0, \tag{2}$$

Since the field A^μ is massless, we demand that the invariant mass satisfies

$$p^2 = p_\mu p^\mu = 0$$

using the four-momentum

$$p^\mu = (E, 0, 0, E)$$

where we see that

$$p_\mu p^\mu = E^2 - E^2 = 0.$$

Then, in order to satisfy Eq. (2), we can only have transverse polarizations

$$\epsilon_{(x)}^\mu = (0, 1, 0, 0)$$

$$\epsilon_{(y)}^\mu = (0, 0, 1, 0)$$

1.2 Vector boson polarization

The $W^\pm$ and Z are the massive vector bosons. Notably, particles with mass have an on-shell condition. In their rest frame, the energy-momentum of a $W^\pm$ or Z boson with mass m is given by

$$p^\mu = (m, 0, 0, 0) \tag{3}$$

and has three spin degrees of freedom, which we can see by applying the gauge fixing identity Eq. (2) as we did for the massless case

$$\epsilon \cdot p = \epsilon^0 m - \epsilon^1 \cdot 0 - \epsilon^2 \cdot 0 - \epsilon^3 \cdot 0 = 0$$

$$\epsilon^0 m = 0$$

$$\epsilon^0 = 0$$

leaving the remaining degrees of freedom

$$\epsilon^\mu = (0, \epsilon^1, \epsilon^2, \epsilon^3).$$

We may then consider a massive vector boson with energy E moving in the z direction, so that

$$p^\mu = (E, 0, 0, p), \quad p_\mu p^\mu = E^2 - p^2 = m^2$$

which means that

$$\epsilon_\mu p^\mu = \epsilon^0 E - \epsilon^3 p = 0$$

$$\epsilon^0 E = \epsilon^3 p$$

$$\boxed{\epsilon^0 = \frac{p}{E} \epsilon^3.}$$

As such, we can define the longitudinal polarization as

$$\begin{aligned} \epsilon_p &= \left(\frac{p}{m}, 0, 0, \frac{E}{m} \right) \\ &= \frac{1}{m} (p, 0, 0, E) \end{aligned}$$

so that we can obtain the ratio

$$\epsilon_0 = \frac{p}{m} = \frac{p}{E} \frac{E}{m} = \frac{p}{E} \epsilon_3.$$

In the ultra-relativistic limit, which we may assume is valid at a high-energy interaction point, we can perform the approximation

$$\begin{aligned}
E^2 &= p^2 + m^2 \\
E &= \sqrt{p^2 + m^2} \\
&= \sqrt{p^2 \left(1 + \frac{m^2}{p^2}\right)} \\
&= p \sqrt{1 + \frac{m^2}{p^2}} \\
&\approx p \sqrt{1 + 0} = p
\end{aligned}$$

when $p \gg m$. Therefore, in this high-energy limit,

$$p_1^\alpha = (E, 0, 0, p) \approx (p, 0, 0, p) \approx (p, 0, 0, E) = m\epsilon_p$$

We may therefore consider the longitudinal spin ϵ_p as proportional to the external momentum p^α

$$\epsilon_p \approx \frac{p^\alpha}{m_1}$$

1.3 Scalar-polarized three-point coupling

Let q_1^α and $\epsilon_{q_1}^\alpha$ define the momentum and polarization of the external particle, respectively. Then, we define the tensor $T_{\beta\gamma}$ by contracting the external polarization with the vertex tensor

$$\begin{aligned}
T_{\beta\gamma} &= \epsilon_{q_1}^\alpha V_{\alpha\beta\gamma} \\
&= \epsilon_{q_1}^\alpha \left[(q_2 - q_3)_\alpha \eta_{\beta\gamma} + (q_3 - q_1)_\beta \eta_{\alpha\gamma} + (q_1 - q_2)_\gamma \eta_{\alpha\beta} \right] \\
&= (\epsilon_{q_1} \cdot (q_2 - q_3)) \eta_{\beta\gamma} + (q_3 - q_1)_\beta \epsilon_{q_1\gamma} + (q_1 - q_2)_\gamma \epsilon_{q_1\beta}
\end{aligned}$$

Incoming scalar polarized leg

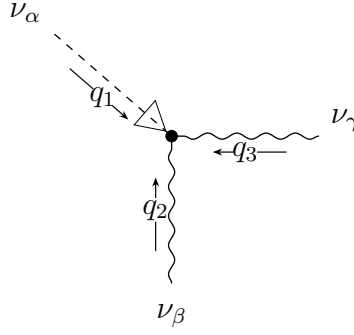


Figure 1.1: Triple-coupling vertex with incoming vector boson ν_α and outgoing vector bosons ν_β and ν_γ . **Future fix:** Adjust the arrow positions so that the momentum arrows are closer to the beginning/ends of the propagator legs. This will leave more visual space for the vertex structure to breathe. Right now the clustered momentum arrows make the structure feel overloaded and crowded.

Then, we assume that our high energy vector boson is scalar polarized. As such, we may make the approximation $\epsilon_{q_1} \approx \frac{q_1}{m}$ so that

$$T_{\beta\gamma} \approx \frac{1}{m_1} (q_1 \cdot (q_2 - q_3)) \eta_{\beta\gamma} + \frac{1}{m_1} (q_3 - q_1)_\beta q_{1\gamma} + \frac{1}{m_1} (q_1 - q_2)_\gamma q_{1\beta}.$$

We may now use momentum conservation $q_1 + q_2 + q_3 = 0$. At the vertex, we see that

$$q_1 = -q_2 - q_3$$

$$\begin{aligned} T_{\beta\gamma} &\approx \frac{1}{m_1} (-q_2 - q_3) \cdot (q_2 - q_3) \eta_{\beta\gamma} + \frac{1}{m_1} (q_2 + 2q_3)_\beta (-q_2 - q_3)_\gamma + \frac{1}{m_1} (-2q_2 - q_3)_\gamma (-q_2 - q_3)_\beta \\ &= \frac{1}{m_1} [(q_3^2 - q_2^2) \eta_{\beta\gamma} + (-q_{2\beta} q_{2\gamma} - q_{2\beta} q_{3\gamma} - 2q_{3\beta} q_{2\gamma} - 2q_{3\beta} q_{3\gamma}) \\ &\quad + (2q_{2\gamma} q_{2\beta} + 2q_{2\gamma} q_{3\beta} + q_{3\gamma} q_{2\beta} + q_{3\gamma} q_{3\beta})] \end{aligned}$$

$$T_{\beta\gamma} \approx \boxed{\frac{1}{m_1} [(q_3^2 - q_2^2) \eta_{\beta\gamma} + q_{2\beta} q_{2\gamma} - q_{3\beta} q_{3\gamma}]}. \quad (4)$$

1.4 Computing the amplitude in unitary gauge

The unitary gauge propagator $D_{\alpha\beta}^{(U)}$ is defined by

$$D_{\alpha\beta}^{(U)} = -i \left(\frac{\eta_{\alpha\beta}^{(H)}}{p^2 - m^2 + i\epsilon} - \frac{p_\alpha p_\beta / m^{2(P)}}{p^2 - m^2 + i\epsilon} \right) \quad (5)$$

$$= \frac{-i \left(\eta_{\alpha\beta} - \frac{p_\alpha p_\beta}{m_\nu^2} \right)}{p^2 - m_\nu^2 + i\epsilon} \quad (6)$$

where TH stands for T'Hooft-Feynman, P stands for Proca, m_ν signifies the mass of a given internal gauge boson. The triple-vertex amplitude can be computed using the formula

$$i\mathcal{M}^{\beta'\gamma'} = iT_{\beta\gamma} D^{\beta\beta'}(q_2) D^{\gamma\gamma'}(q_3). \quad (7)$$

We may then compute the amplitude by inserting the contracted polarized vertex tensor Eq. (4) and the unitary gauge propagators as denoted in Eq. (6) into Eq. (7)

$$\begin{aligned} \mathcal{M}^{\beta'\gamma'} &= T_{\beta\gamma} \frac{-i}{q_2^2 - m^2 + i\epsilon} \left(\eta^{\beta\beta'} - \frac{q_2^\beta q_2^{\beta'}}{m^2} \right) \frac{-i}{q_3^2 - m^2 + i\epsilon} \left(\eta^{\gamma\gamma'} - \frac{q_3^\gamma q_3^{\gamma'}}{m^2} \right) \\ &= \frac{-1}{(q_2^2 - m^2 + i\epsilon)(q_3^2 - m^2 + i\epsilon)} T_{\beta\gamma} \left(\eta^{\beta\beta'} - \frac{q_2^\beta q_2^{\beta'}}{m^2} \right) \left(\eta^{\gamma\gamma'} - \frac{q_3^\gamma q_3^{\gamma'}}{m^2} \right). \end{aligned}$$

We continue to simplify by contracting $T_{\beta\gamma}$ with the propagator $D^{\beta\beta'}(q_2)$ so that

$$\begin{aligned} \mathcal{M}^{\beta'\gamma'} &= \frac{-1}{(q_2^2 - m^2 + i\epsilon)(q_3^2 - m^2 + i\epsilon)} \left(T_{\beta\gamma} \eta^{\beta\beta'} - T_{\beta\gamma} \frac{q_2^\beta q_2^{\beta'}}{m^2} \right) \left(\eta^{\gamma\gamma'} - \frac{q_3^\gamma q_3^{\gamma'}}{m^2} \right) \\ &= \frac{-1}{(q_2^2 - m^2 + i\epsilon)(q_3^2 - m^2 + i\epsilon)} \left(T_\gamma^{\beta'} - \frac{q_2^{\beta'}}{m^2} q_2^\beta T_{\beta\gamma} \right) \left(\eta^{\gamma\gamma'} - \frac{q_3^\gamma q_3^{\gamma'}}{m^2} \right). \end{aligned}$$

We then expand the expression and contract it with the $D^{\gamma\gamma'}(q_3)$ propagator

$$\begin{aligned}
\mathcal{M}^{\beta'\gamma'} &= \frac{-1}{(q_2^2 - m^2 + i\epsilon)(q_3^2 - m^2 + i\epsilon)} \left(T_\gamma^{\beta'} - \frac{q_2^{\beta'}}{m^2} q_2^\beta T_{\beta\gamma} \right) \left(\eta^{\gamma\gamma'} - \frac{q_3^\gamma q_3^{\gamma'}}{m^2} \right) \\
&= \frac{-1}{(q_2^2 - m^2 + i\epsilon)(q_3^2 - m^2 + i\epsilon)} \left[T_\gamma^{\beta'} \eta^{\gamma\gamma'} - T_\gamma^{\beta'} q_3^\gamma \frac{q_3^{\gamma'}}{m^2} - \frac{q_2^{\beta'}}{m^2} q_2^\beta T_{\beta\gamma} \eta^{\gamma\gamma'} \right. \\
&\quad \left. + \frac{q_2^{\beta'}}{m^2} q_2^\beta T_{\beta\gamma} q_3^\gamma \frac{q_3^{\gamma'}}{m^2} \right] \\
&= \frac{-1}{(q_2^2 - m^2 + i\epsilon)(q_3^2 - m^2 + i\epsilon)} \left[T^{\beta'\gamma'}_\gamma - \frac{q_3^{\gamma'}}{m^2} q_3^\gamma T_\gamma^{\beta'} - \frac{q_2^{\beta'}}{m^2} q_2^\beta T_{\beta}^{\gamma'} + \frac{q_2^{\beta'} q_3^{\gamma'}}{m^4} q_2^\beta q_3^\gamma T_{\beta\gamma} \right]
\end{aligned}$$

Now, we may evaluate the contracted objects

$$\begin{aligned}
q_3^\gamma T_\gamma^{\beta'} &= q_3^\gamma \frac{[(q_3^2 - q_2^2) \eta_\gamma^{\beta'} + q_{2\gamma} q_2^{\beta'} - q_{3\gamma} q_3^{\beta'}]}{m} \\
&= \frac{1}{m} \left[(q_3^2 - q_2^2) q_3^{\beta'} + (q_2 \cdot q_3) q_2^{\beta'} - q_3^2 q_3^{\beta'} \right] \\
&= \frac{1}{m} \left[q_3^2 q_3^{\beta'} - q_2^2 q_3^{\beta'} + (q_2 \cdot q_3) q_2^{\beta'} - q_3^2 q_3^{\beta'} \right] \\
&= \frac{1}{m} \left(-q_2^2 q_3^{\beta'} + (q_2 \cdot q_3) q_2^{\beta'} \right)
\end{aligned}$$

$$\begin{aligned}
q_2^\beta T_\beta^{\gamma'} &= \frac{q_2^\beta}{m} \left[(q_3^2 - q_2^2) \eta_\beta^{\gamma'} + q_{2\beta} q_2^{\gamma'} - q_{3\beta} q_3^{\gamma'} \right] \\
&= \frac{1}{m} \left[(q_3^2 - q_2^2) q_2^{\gamma'} + q_2^2 q_2^{\gamma'} - (q_2 \cdot q_3) q_3^{\gamma'} \right] \\
&= \frac{1}{m} \left[q_3^2 q_2^{\gamma'} - q_2^2 q_2^{\gamma'} + q_2^2 q_2^{\gamma'} - (q_2 \cdot q_3) q_3^{\gamma'} \right] \\
&= \frac{1}{m} \left(q_3^2 q_2^{\gamma'} - (q_2 \cdot q_3) q_3^{\gamma'} \right)
\end{aligned}$$

$$\begin{aligned}
\frac{q_2^{\beta'} q_3^{\gamma'}}{m^4} q_2^\beta q_3^\gamma T_{\beta\gamma} &= \frac{q_2^{\beta'} q_3^{\gamma'}}{m^4} q_2^\beta \frac{q_3^\gamma}{m} [(q_3^2 - q_2^2) \eta_{\beta\gamma} + q_{2\beta} q_{2\gamma} - q_{3\beta} q_{3\gamma}] \\
&= \frac{q_2^{\beta'} q_3^{\gamma'}}{m^4} q_2^\beta \frac{1}{m} (-q_2^2 q_{3\beta} + (q_2 \cdot q_3) q_{2\beta}) \\
&= \frac{q_2^{\beta'} q_3^{\gamma'}}{m^4} \frac{1}{m} (-(q_2 \cdot q_3) q_2^2 + (q_2 \cdot q_3) q_2^2) \\
&= 0
\end{aligned}$$

Thus, we get that

$$\begin{aligned}
\mathcal{M}^{\beta'\gamma'} &= \frac{-1}{(q_2^2 - m_\nu^2 + i\epsilon) \cdot (q_3^2 - m_\nu^2 + i\epsilon)} \cdot \\
&\frac{1}{m} \left[\left((q_3^2 - q_2^2) \eta^{\beta'\gamma'} + q_2^{\beta'} q_2^{\gamma'} - q_3^{\beta'} q_3^{\gamma'} \right) \right. \\
&\quad \left. - \underbrace{\frac{q_3^{\gamma'}}{m_\nu^2} (-q_2^2 q_3^{\beta'} + (q_2 \cdot q_3) q_2^{\beta'})}_{(a)} - \underbrace{\frac{q_2^{\beta'}}{m_\nu^2} (q_3^2 q_2^{\gamma'} - (q_2 \cdot q_3) q_3^{\gamma'})}_{(b)} \right].
\end{aligned}$$

Next, we simplify each term further

$$\begin{aligned}
(a) &= \frac{q_3^{\gamma'}}{m_\nu^2} (-q_2^2 q_3^{\beta'} + (q_2 \cdot q_3) q_2^{\beta'}) \\
&= \frac{-q_2^2 q_3^{\beta'} q_3^{\gamma'}}{m_\nu^2} + \frac{(q_2 \cdot q_3) q_2^{\beta'} q_3^{\gamma'}}{m_\nu^2}
\end{aligned}$$

$$\begin{aligned}
(b) &= \frac{q_2^{\beta'}}{m_\nu^2} (q_3^2 q_2^{\gamma'} - (q_2 \cdot q_3) q_3^{\gamma'}) \\
&= \frac{q_3^2 q_2^{\beta'} q_2^{\gamma'}}{m_\nu^2} - \frac{(q_2 \cdot q_3) q_3^{\gamma'} q_2^{\beta'}}{m_\nu^2}
\end{aligned}$$

so that we get

$$\begin{aligned}\mathcal{M}^{\beta'\gamma'} &= \frac{-1}{(q_2^2 - m_\nu^2 + i\epsilon) \cdot (q_3^2 - m_\nu^2 + i\epsilon)} \cdot \\ &\frac{1}{m} \left[\left((q_3^2 - q_2^2) \eta^{\beta'\gamma'} + q_2^{\beta'} q_2^{\gamma'} - q_3^{\beta'} q_3^{\gamma'} \right) \right. \\ &\quad \left. + \frac{q_2^2 q_3^{\beta'} q_3^{\gamma'}}{m_\nu^2} - \frac{(q_2 \cdot q_3) q_2^{\beta'} q_3^{\gamma'}}{m_\nu^2} - \frac{q_3^2 q_2^{\beta'} q_2^{\gamma'}}{m_\nu^2} + \frac{(q_2 \cdot q_3) q_3^{\gamma'} q_2^{\beta'}}{m_\nu^2} \right]\end{aligned}$$

which gives us

$$\begin{aligned}\mathcal{M}^{\beta'\gamma'} &= \frac{-1}{(q_2^2 - m_\nu^2 + i\epsilon) (q_3^2 - m_\nu^2 + i\epsilon)} \cdot \\ &\frac{1}{m} \left[\left((q_3^2 - q_2^2) \eta^{\beta'\gamma'} + q_2^{\beta'} q_2^{\gamma'} - q_3^{\beta'} q_3^{\gamma'} \right) + \frac{q_2^2 q_3^{\beta'} q_3^{\gamma'}}{m_\nu^2} - \frac{q_3^2 q_2^{\beta'} q_2^{\gamma'}}{m_\nu^2} \right] \\ &= \frac{-1}{(q_2^2 - m_\nu^2 + i\epsilon) (q_3^2 - m_\nu^2 + i\epsilon)} \cdot \\ &\frac{1}{m} \left[(q_3^2 - q_2^2) \eta^{\beta'\gamma'} + \left(1 - \frac{q_3^2}{m_\nu^2} \right) q_2^{\beta'} q_2^{\gamma'} - \left(1 - \frac{q_2^2}{m_\nu^2} \right) q_3^{\beta'} q_3^{\gamma'} \right] \\ &= \frac{-1}{(q_2^2 - m_\nu^2 + i\epsilon) (q_3^2 - m_\nu^2 + i\epsilon)} \cdot \\ &\frac{1}{m} \left[(q_3^2 - q_2^2) \eta^{\beta'\gamma'} + \frac{m_\nu^2 - q_3^2}{m_\nu^2} q_2^{\beta'} q_2^{\gamma'} - \frac{m_\nu^2 - q_2^2}{m_\nu^2} q_3^{\beta'} q_3^{\gamma'} \right] \\ &= \frac{-1}{(q_2^2 - m_\nu^2 + i\epsilon) (q_3^2 - m_\nu^2 + i\epsilon)} \cdot \\ &\frac{1}{m} \left[(q_3^2 - q_2^2) \eta^{\beta'\gamma'} - \frac{q_3^2 - m_\nu^2}{m_\nu^2} q_2^{\beta'} q_2^{\gamma'} + \frac{q_2^2 - m_\nu^2}{m_\nu^2} q_3^{\beta'} q_3^{\gamma'} \right].\end{aligned}$$

Now, we define the propagator denominators

$$\boxed{D_{(q_2)} = q_2^2 - m_\nu^2 + i\epsilon, \quad D_{(q_3)} = q_3^2 - m_\nu^2 + i\epsilon}$$

so that we get the amplitude

$$\begin{aligned}
\mathcal{M}^{\beta'\gamma'} &= -\frac{1}{m} \cdot \left[\frac{(q_3^2 - q_2^2) \eta^{\beta'\gamma'}}{D_{(q_2)} D_{(q_3)}} - \frac{\frac{D_{(q_3)}}{m_\nu^2} q_2^{\beta'} q_2^{\gamma'}}{D_{(q_2)} D_{(q_3)}} + \frac{\frac{D_{(q_2)}}{m_\nu^2} q_3^{\beta'} q_3^{\gamma'}}{D_{(q_2)} D_{(q_3)}} \right] \\
&= -\frac{1}{m} \cdot \left[\frac{\eta^{\beta'\gamma'} [(q_3^2 - m_\nu^2) - (q_2^2 - m_\nu^2)]}{D_{(q_2)} D_{(q_3)}} - \frac{D_{(q_3)} \frac{q_2^{\beta'} q_2^{\gamma'}}{m_\nu^2}}{D_{(q_2)} D_{(q_3)}} + \frac{D_{(q_2)} \frac{q_3^{\beta'} q_3^{\gamma'}}{m_\nu^2}}{D_{(q_2)} D_{(q_3)}} \right] \\
&= -\frac{1}{m} \cdot \left[\frac{\eta^{\beta'\gamma'} [D_{(q_3)} - D_{(q_2)}]}{D_{(q_2)} D_{(q_3)}} - \frac{D_{(q_3)} \frac{q_2^{\beta'} q_2^{\gamma'}}{m_\nu^2}}{D_{(q_2)} D_{(q_3)}} + \frac{D_{(q_2)} \frac{q_3^{\beta'} q_3^{\gamma'}}{m_\nu^2}}{D_{(q_2)} D_{(q_3)}} \right]
\end{aligned}$$

which we simplify to obtain the difference of two propagators in the final form

$$\begin{aligned}
\mathcal{M}^{\beta'\gamma'} &= -\frac{1}{m} \cdot \left[\frac{\eta^{\beta'\gamma'}}{D_{(q_2)}} - \frac{\eta^{\beta'\gamma'}}{D_{(q_3)}} - \frac{\frac{q_2^{\beta'} q_2^{\gamma'}}{m_\nu^2}}{D_{(q_2)}} + \frac{\frac{q_3^{\beta'} q_3^{\gamma'}}{m_\nu^2}}{D_{(q_3)}} \right] \\
&= \frac{1}{m} \cdot \left[-\frac{\eta^{\beta'\gamma'}}{D_{(q_2)}} + \frac{\eta^{\beta'\gamma'}}{D_{(q_3)}} + \frac{\frac{q_2^{\beta'} q_2^{\gamma'}}{m_\nu^2}}{D_{(q_2)}} - \frac{\frac{q_3^{\beta'} q_3^{\gamma'}}{m_\nu^2}}{D_{(q_3)}} \right]
\end{aligned}$$

$$\boxed{i\mathcal{M}^{\beta'\gamma'} = \frac{i}{m_{\text{in}}} \cdot \left[\frac{\eta^{\beta'\gamma'} - \frac{q_3^{\beta'} q_3^{\gamma'}}{m_\nu^2}}{D_{(q_3)}} - \frac{\eta^{\beta'\gamma'} - \frac{q_2^{\beta'} q_2^{\gamma'}}{m_\nu^2}}{D_{(q_2)}} \right]}$$

2 THE WARD IDENTITY

2.1 the baby identity we want to prove

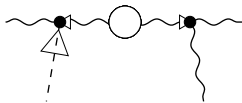


Figure 2.1: Box one-loop $W^+W^- \rightarrow W^+W^-$ topology.

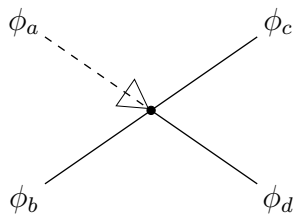


Figure 2.2: Box one-loop $W^+W^- \rightarrow W^+W^-$ topology.

2.2 the big identity we want to prove

3 ONE LOOP $W^+W^- \rightarrow W^+W^-$ DIAGRAMS

In each one-loop diagram, let the four external momenta be defined by q_1 , q_2 , q_3 , and q_4 with momentum conserved at each vertex. We must integrate over the momentum on each side of the box under one integrand since the momenta do not factorize.

3.1 Box diagram

The box is the first possible $W^+W^- \rightarrow W^+W^-$ one-loop topology.

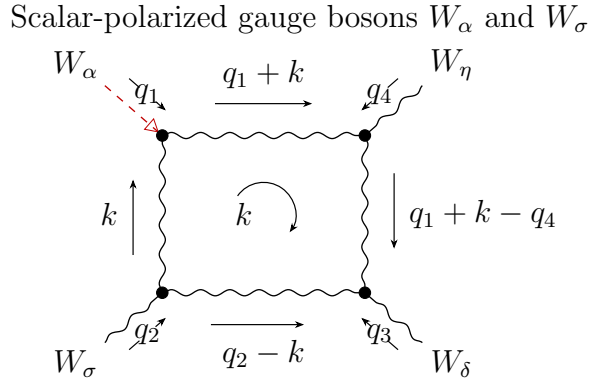


Figure 3.1: Box one-loop $W^+W^- \rightarrow W^+W^-$ topology.

3.2 Cross-box diagram

The cross-box is the second possible $W^+W^- \rightarrow W^+W^-$ one-loop topology.

3.3 Triangle-box diagram

The triangle-box is the third possible $W^+W^- \rightarrow W^+W^-$ one-loop topology.

4 $SU(2)$ SUBGROUP

1. Right now, we are not including the photon and are only consider the subgroup $SU(2)$

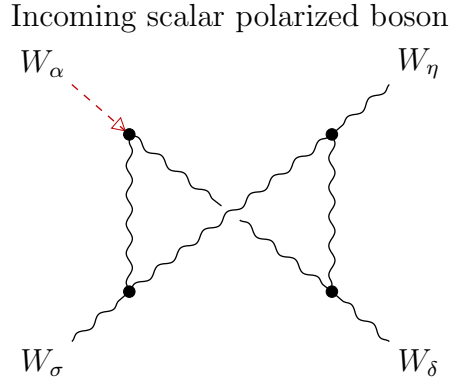


Figure 3.2: Cross-box one-loop $W^+W^- \rightarrow W^+W^-$ topology

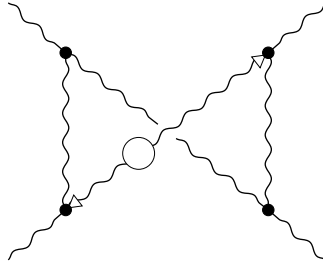


Figure 3.3: Cross-box one-loop $W^+W^- \rightarrow W^+W^-$ topology

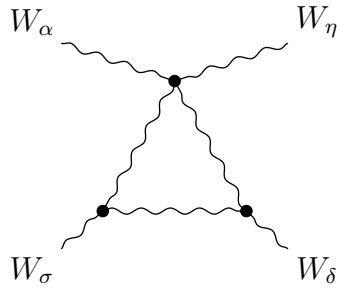


Figure 3.4: Triangle-box one-loop $W^+W^- \rightarrow W^+W^-$ topology

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2. Considering just the subgroup $SU(2)$ means that we are (essentially) only left with the W^+, W^- and Z bosons
 3. This means that we are left with two step operators and one neutral operator
 4. Typically, the photon γ , represented by the photon field A_μ , accounts for the extra degree of freedom provided by the $U(1)$ in the total gauge group $SU(2) \times U(1)$